

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built a SaaS platform to automate research & case-writing in competitive high school debate, reaching **10K+ users**
- Designed a text retrieval algorithm that extracts evidence segments from a debate case using URL validation & IQR filtering. Indexed **600K+ documents** on an *Elasticsearch* engine, returning query results in **less than 1 second**
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat
- Integrated system scalability with an automated pipeline that pushes weekly updates of **1,000+ new documents**

Product Development Intern | Jetson

January 2025 – February 2025

- Collaborated on **7 iterations of UI/UX testing** across 3 experimental features of the Jetson App (100K+ users)
- Communicated ideas with candor in a **5-page report** recommending ways to optimize engagement for Jetson's CEO
- Redesigned onboarding in *Figma*, cutting it from **12 to 5 pages**, and validated it via **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)* and *Microsoft Excel* to find outreach patterns
- Identified that “science fun facts” posts drew 69.3% of views despite making up only 7% of all content posted
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

PROJECTS

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Designed a **5-step Python scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking, validated with **14 unit and integration test cases**
- Built a **scheduling agent** (*React/Firebase*) that auto-books meetings for large groups (30+ ppl) on Google Calendar
- Created a lobby system that lets the host authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a team of **4** on a pricing algorithm for **Kamion (YC S22)**. **2nd Place at UIUC Founder's Hackathon**
- Leveraged *Claude Sonnet 4.6* as an AI tool: wrote a *JSON schema* for an *agent* to extract condition, brand, model and year from photos, fine-tuned the model after each wrong prediction until **10 consecutive images** were correct
- Built a *quantile regression* pricing model on the extracted features, reaching **87% accuracy** on **30 samples**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built an interactive citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**
- Created a dual-API ingestion pipeline resolving papers via *OpenAlex* & backfilling metadata from *Semantic Scholar*
- Deployed a **RAG agent** (*Claude Sonnet 4.5*) that retrieves session-scoped vector documents (*Supernemory*)
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count

SKILLS

Programming Languages: Python, Java, JavaScript, C++, SQL, HTML/CSS

Frameworks/Platforms: FastAPI, Flask, React, Next.js, Node.js, PostgreSQL, AWS, Firebase, Supabase, Vercel

DevTools: Git, GitHub, Claude Code, Codex, VS Code, Docker, CI/CD, Figma, Bun, Catch2, Clang

AI/ML PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, AI Agents, AI Automation, LLMs, RAG

Concepts/Practices: Object Oriented Programming, Prompt Engineering, GenAI, Algorithms, AI Tools